

DustTwin AI Model: Now & Next

The trained PM10 model is ready to connect to your frontend. Advanced AI features remain future work.

CURRENT FORECAST PIPELINE

1. PM10 HISTORY
Past 120 seconds / 121 snapshots
Laboratory OPC-N3 data. Wind and site control belong to a separate software simulation.

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2. TRAINED AI MODEL
HistGBM / 16 PM10 features

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3. PM10 FORECAST
Predicted value at +30 seconds

1. AI AVAILABLE NOW

Implemented and verified

Real trained model

The fitted HistGBM model is included in Git. No retraining is needed to use it.

30-second PM10 forecast

Uses 120 seconds of past PM10. It rejects missing, stale or future input.

Live API + recorded demo

Live inference and six measured recordings are ready. Saved replay is clearly labelled.

Values for your dashboard

Returns current and predicted PM10, two simple baselines and input details.

Later prediction checking

Recorded actual is revealed only when the earlier forecast reaches its target time.

Verified handoff

Works from a fresh clone. If the live model is unavailable, requests return an error.

2. CURRENT LIMITS

Keep these clear in the demo

PM10 only

No learned prediction of PM2.5, wind, humidity or temperature. Field sensor streaming is not integrated.

A point forecast

No calibrated confidence or exact crossing ETA. A 30-second horizon does not guarantee 30 seconds of warning.

Laboratory evidence

Outdoor boundary accuracy, physical misting and real water savings have not been validated.

Separate site simulation

Configured wind and layout feed rule-based zone decisions. Zone commands and water use are simulated.

Your integration step

Connect your existing frontend to the API. An online website also needs a Python host; GitHub alone does not run it.

3. FUTURE AI WORK

Proposed; not implemented

New field data

Collect independent site/sensor data. Reserve a new, untouched evaluation set.

More measured inputs

Train with PM2.5, wind, humidity and temperature once aligned data exists.

Uncertainty + sensor faults

Test forecast intervals, crossing probabilities and sensor-quality checks on new data.

Plume + misting models

Explore physics-informed AI and droplet efficiency with measured site/actuator data.

Learned spray control

Research reinforcement learning to balance exposure and water in a validated environment.

Edge + controlled updates

Assess lightweight deployment and monitored, versioned retraining with rollback.

MEASURED LABORATORY RESULTS

Forecast	MAE	RMSE
Trained AI	88.405	179.272
Last reading	95.702	199.384
Trailing mean	81.565	200.713

Errors in $\mu\text{g}/\text{m}^3$. MAE = average absolute error. AI beats the last reading on MAE; the trailing mean is better.

VERIFICATION & WARNING LIMITS

27,178 saved forecasts rechecked.

28 Python tests + 3 browser checks pass.

Test: 15,065 overlapping windows in only 3 related lab recordings.

Warning check: 13/18 matched runs, 15 false alerts; only 1/3 first onsets warned.